



PROGRAMMABLE LOGIC CONTROLLERS

A Practical Course Guide

From Fundamentals to Industrial Application

Prepared by

Awet G. Nway

BSc Mechanical Engineering · Robotics & PLC Specialist

Founder, AYE Tech Hub

Engineering the Future through Technology & Knowledge



Table of Contents

1. About This Course.....	3
2. Introduction — What is a PLC?.....	3
3. History & Evolution of the PLC.....	5
4. PLC Block Diagram & Components.....	6
5. How a PLC Works — The Scan Cycle.....	8
6. Types of PLCs — Compact vs Modular.....	10
7. PLC Programming Languages (IEC 61131-3).....	12
8. Ladder Logic — Rungs, Rails & Basic Instructions.....	13
9. Timers, Counters & Math Instructions.....	15
10. PLC I/O — Sinking and Sourcing.....	17
11. Sensors Used with PLCs.....	18
12. Communication Protocols.....	19
13. Major PLC Brands & Manufacturers.....	20
14. Applications of PLCs in Industry.....	21
15. Advantages & Disadvantages of PLCs.....	22
16. Safety, Redundancy & Best Practices.....	23
17. Troubleshooting & Maintenance.....	24
18. Career Pathways in PLC & Automation.....	25
19. Learning Roadmap & Resources.....	26
20. Final Words from AYE Tech Hub.....	27



1. About This Course

This document is a comprehensive course on Programmable Logic Controllers — the backbone of modern industrial automation. It is designed for engineering students, technicians transitioning into automation, maintenance engineers, and self-learners who want a clear and practical path from PLC basics to industrial application.

No prior programming experience is required. By the end of this guide, you will understand what a PLC is, what it is made of, how it works internally, how it is programmed, the different communication protocols used in the field, and how to apply this knowledge in real factories and plants.

Who this course is for

Engineering students (mechanical, electrical, mechatronics, electronics), technicians, maintenance and production team leaders, automation engineers, and anyone preparing for an interview or a job involving PLCs and SCADA systems.

2. Introduction — What is a PLC?

A Programmable Logic Controller, or PLC, is a solid-state, computerized industrial controller that performs discrete or sequential logic in a factory or automation environment. Think of it as a ruggedized industrial computer specifically designed to run continuously in harsh conditions — heat, dust, vibration, electrical noise — while reliably controlling machines and processes.

Formal definition (NEMA)

The National Electrical Manufacturers Association (NEMA) defines a PLC as: a digital electronic device that uses programmable memory to store instructions and to implement specific functions such as logic, sequencing, timing, counting, and arithmetic, in order to control machines and processes.

Why PLCs replaced relay panels

Before PLCs, industrial control was done with banks of electromechanical relays — hundreds or thousands of physical components wired together. Changing the logic meant rewiring the panel. PLCs replaced that hardware with software: the same control logic now lives as code inside a small box, and changing it takes a few minutes on a laptop instead of days with wire cutters.

- **Reliable** — Designed to run 24/7 in industrial environments for years.
- **Flexible** — Logic changes are software changes, not rewiring.
- **Compact** — One small unit replaces large relay panels.
- **Networkable** — Can talk to SCADA, HMIs, robots, and other PLCs.



-
- **Diagnosable** — Built-in fault detection and status indicators.



3. History & Evolution of the PLC

The PLC was born from a real industrial problem. In the late 1960s, General Motors faced enormous costs every time it had to change the relay logic on its assembly lines. They issued a famous specification asking for a programmable replacement for relay banks — one that could be reprogrammed without rewiring.

The father of the PLC

In 1968, Richard E. "Dick" Morley and his team at Bedford Associates designed the very first PLC, the Modicon 084, to meet that GM specification. The first unit was installed at a General Motors plant in 1969, and the modern automation industry was born. Dick Morley is widely regarded as the father of the PLC.

Generations of PLCs

Era	Key Development
1968–1975	First-generation PLCs replace relay logic. Limited memory, basic ladder logic.
1975–1985	Faster CPUs, more memory, analog I/O, math functions, remote I/O.
1985–1995	Networking and fieldbuses appear (Modbus, Profibus). PLCs talk to HMIs and SCADA.
1995–2010	Ethernet integration, motion control, web servers built into PLCs.
2010–today	Industry 4.0: smart PLCs, IIoT connectivity, cloud integration, edge computing.

Why this history matters

Understanding where the PLC came from helps you understand why it still uses ladder logic — that graphical language was designed to look familiar to engineers who used to read relay schematics. That backward compatibility is a key reason PLCs have survived for over 50 years.

4. PLC Block Diagram & Components

Every PLC, no matter the brand or size, is built from the same fundamental building blocks. Understanding these components and how they connect is the foundation of all PLC work — from programming to wiring to troubleshooting.

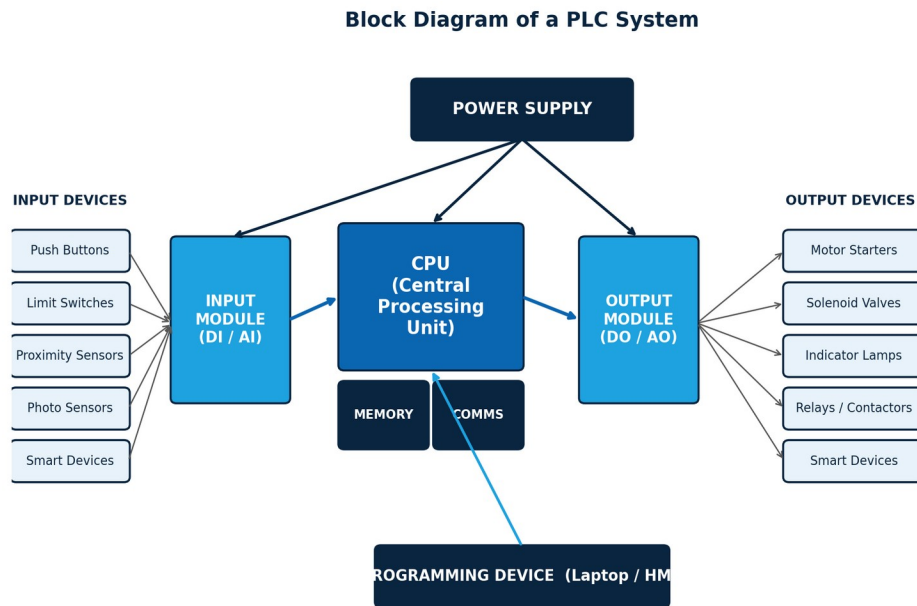


Figure 4.1 — Block diagram of a typical PLC system showing the major components and signal flow.

4.1 Input / Output (I/O) Modules

The I/O modules are the eyes and hands of the PLC. Input modules read signals from field devices like push buttons, limit switches, and proximity sensors. Output modules drive devices like motor starters, solenoid valves, lamps, and contactors. I/O can be digital (on/off) or analog (continuous values like temperature, pressure, or flow rate).

- **Digital Input (DI)** — Reads discrete on/off signals (24 V DC or 110/220 V AC).
- **Digital Output (DO)** — Switches discrete loads on and off (relay, transistor, or triac type).
- **Analog Input (AI)** — Reads continuous signals like 4-20 mA or 0-10 V from sensors.
- **Analog Output (AO)** — Sends continuous signals to variable speed drives or control valves.

4.2 Power Supply

The power supply converts incoming AC line voltage (usually 220 V AC or 110 V AC) or 24 V DC into the regulated DC voltages required internally by the CPU and I/O modules. Power supplies are sized based on the total current draw of the rack. A correctly sized power supply with a small safety margin is essential — undersizing causes intermittent faults that are very hard to diagnose.



4.3 Central Processing Unit (CPU)

The CPU is the brain of the PLC. It executes the user program stored in memory, performs all arithmetic and logic operations, controls communications, and manages the I/O. The CPU's scan time — how fast it completes one full cycle of reading inputs, executing logic, and updating outputs — is one of the key performance parameters of any PLC.

A modern industrial CPU contains a microprocessor, system memory, user memory, communication processors, and supervisory circuits that monitor its own health. If anything goes wrong internally, the CPU sets diagnostic bits that the programmer can read.

4.4 Memory System

PLC memory is organized into several distinct areas, each with a specific purpose.

Memory Area	Purpose
Input Image	Mirrors the current state of all physical inputs at the start of each scan.
Output Image	Holds the desired state of all outputs to be written at the end of each scan.
Data / Variable	User-defined variables: timers, counters, integers, real numbers, strings.
User Program	The ladder logic, function blocks, or structured text written by the programmer.
System / Executive	Firmware that runs the PLC itself (protected, not user-accessible).

Memory types include RAM (fast, volatile working memory), Flash or EEPROM (non-volatile storage that retains the program when power is removed), and battery-backed RAM in older PLCs for retentive data.

4.5 Communication Interface

Modern PLCs include one or more communication ports, allowing them to talk to programming computers, HMIs, SCADA systems, drives, and other PLCs. Common interfaces include Ethernet, RS-485, RS-232, USB, and proprietary fieldbus connectors. We cover protocols in detail in Section 12.

4.6 Programming Device

The programming device is normally a laptop or PC running the manufacturer's software (Siemens TIA Portal, Rockwell Studio 5000, Mitsubishi GX Works, and so on). The laptop connects to the PLC over USB, serial, or Ethernet. Once the program is downloaded and the PLC is running, the laptop can be disconnected — the program runs from the PLC's own memory.

5. How a PLC Works — The Scan Cycle

The single most important concept in PLC operation is the scan cycle. A PLC does not execute its program once — it executes it continuously, in a loop, thousands of times per second. Understanding this loop is the difference between guessing and knowing how your program will behave.

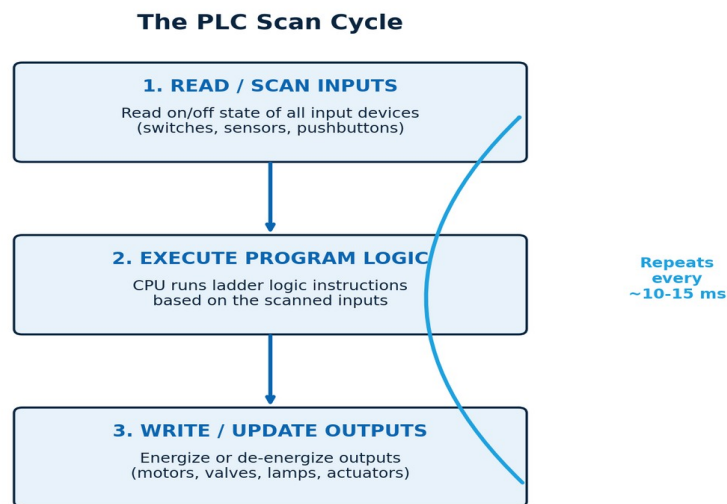


Figure 5.1 — The PLC scan cycle. The cycle repeats continuously, typically every 10–15 ms.

Step 1 — Read / Sense the Inputs

At the start of every scan, the PLC samples the on/off state of every physical input terminal and stores the result in the Input Image table in memory. From this point on, the program does not look at the physical inputs again during this scan — it reads only the snapshot stored in memory.

Inputs being scanned include pushbuttons, selector switches, limit switches, proximity sensors, photoelectric sensors, pressure switches, level switches, and analog signals from temperature or flow transmitters.

Step 2 — Execute the Program Logic

The CPU now executes the user program one rung (or one instruction block) at a time, from top to bottom and left to right. For each instruction, it looks up the current value from the Input Image, performs the logic, and writes the result to the Output Image (or to internal memory bits, timers, or counters).

Crucially, the physical outputs are not yet updated. Only the Output Image table in memory is being modified during this step. This is why output signals only change at the end of the scan, not the moment the logic decides them.



Step 3 — Write / Update the Outputs

Finally, the PLC copies the entire Output Image table to the physical output terminals in one operation. Motors energize, valves shift, lamps light up, and actuators move. Then the cycle starts again at Step 1.

Scan time and why it matters

A typical industrial PLC completes one full scan in 10 to 15 milliseconds, though high-end CPUs can run sub-millisecond scans. Scan time depends on three factors: the speed of the CPU, the size of the user program, and the number of I/O points being scanned. For fast processes like high-speed packaging, scan time directly limits how fast the machine can react — so designers must minimize program length and use techniques like interrupts or high-speed counters for critical signals.

A critical pitfall for beginners

Because the PLC only reads inputs once per scan, very short pulses — shorter than the scan time — can be missed entirely. If you need to catch a fast signal (an encoder pulse, a high-speed photoelectric trigger), you must use a hardware high-speed input or interrupt, not a normal input.

6. Types of PLCs — Compact vs Modular

PLCs come in two main physical configurations: compact (fixed) and modular (rack-mounted). The choice between them depends on the size of the project, future expansion plans, and the budget.

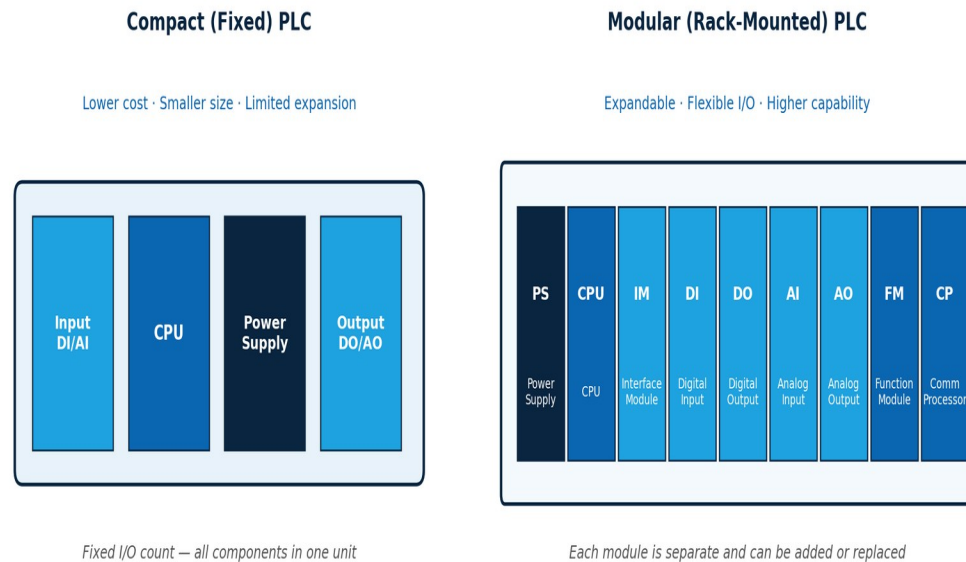


Figure 6.1 — Compact (fixed) PLC versus modular (rack-mounted) PLC.

6.1 Compact PLC

A compact PLC, also called an integrated or fixed PLC, has all its components — power supply, CPU, and a set number of inputs and outputs — built into a single housing. Examples include the Siemens LOGO!, Allen-Bradley Micro800 series, and Mitsubishi FX series.

- **Best for** — Small machines and projects with up to ~30 I/O points.
- **Strengths** — Low cost, very compact, easy to install, fast to commission.
- **Limitations** — Fixed I/O count. Limited expansion options. If you need more inputs, you must upgrade the whole unit.

6.2 Modular PLC

A modular PLC consists of separate plug-in modules mounted on a rack or chassis: a power supply module, a CPU module, then a flexible mix of digital input, digital output, analog input, analog output, communication, and specialty modules. You add or replace modules as your project grows.

- **Best for** — Medium to large machines, complete production lines, plant-wide systems.
- **Strengths** — Highly expandable, mix-and-match I/O, hot-swappable modules in many systems, distributed I/O over networks.



- **Limitations** — Higher initial cost, larger footprint, requires proper rack design.

6.3 Comparison summary

Aspect	Compact PLC	Modular PLC
I/O Count	Fixed (typically 6–30 points)	Expandable (hundreds to thousands)
Cost	Lower	Higher
Size	Very small	Larger (rack-based)
Flexibility	Limited	High — add modules as needed
Typical Use	Small machines, OEM equipment	Production lines, large plants
Example Models	Siemens LOGO!, AB Micro800	Siemens S7-1500, AB ControlLogix

6.4 Other classifications

PLCs can also be classified by size — nano, micro, small, medium, large — based on I/O capacity and CPU power. Another modern category is the soft PLC, which runs the PLC engine in software on a standard industrial PC, often used in motion control and complex machine-vision applications.

7. PLC Programming Languages (IEC 61131-3)

The international standard IEC 61131-3 defines five programming languages for PLCs. Modern engineering software like Siemens TIA Portal and Rockwell Studio 5000 supports several of these, and you can even mix them within a single project.

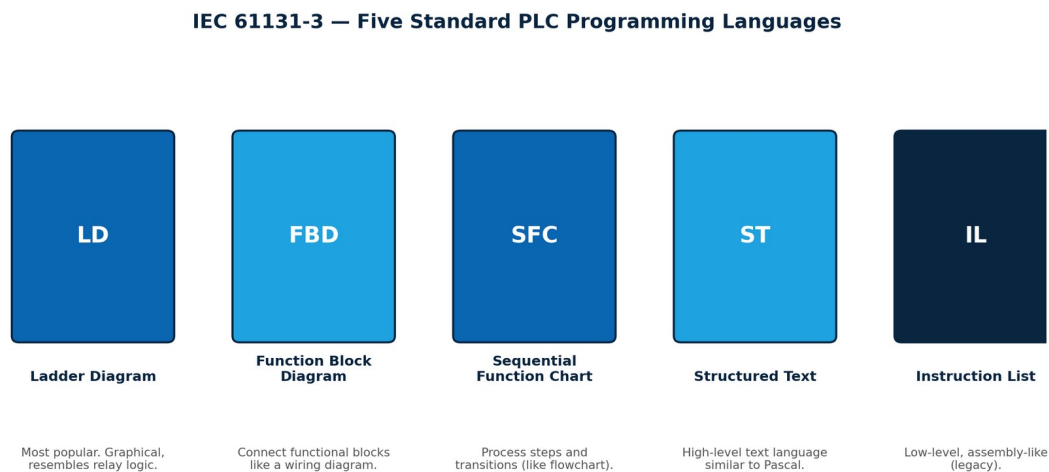


Figure 7.1 — The five standard PLC programming languages defined by IEC 61131-3.

- **Ladder Diagram (LD)** — Graphical, looks like relay schematics. The most widely used language; it is the natural choice for electricians and maintenance teams.
- **Function Block Diagram (FBD)** — Graphical. Blocks are wired together like an electronic schematic. Excellent for analog control, PID loops, and signal processing.
- **Sequential Function Chart (SFC)** — Graphical. Represents a process as steps and transitions, similar to a flowchart. Ideal for state-machine processes like batch control.
- **Structured Text (ST)** — Text-based, similar to Pascal. Powerful for math, loops, and complex algorithms that would be awkward in ladder logic.
- **Instruction List (IL)** — Text-based, low-level, assembly-like. Largely deprecated in modern systems but still seen on older Siemens platforms (STL).

Which language should you learn first?

Start with Ladder Diagram. It is the most common in industry, the easiest to read for maintenance staff, and works on every PLC platform. Once you are comfortable with LD, add Function Block Diagram for analog control, then Structured Text for advanced logic.

8. Ladder Logic — Rungs, Rails & Basic Instructions

Ladder logic gets its name because it looks like a ladder: two vertical power rails on either side, with horizontal rungs between them carrying the control logic. The left rail is the positive power supply, the right rail is the return. Each rung represents one logical equation: if the conditions on the left are true, the output on the right is energized.

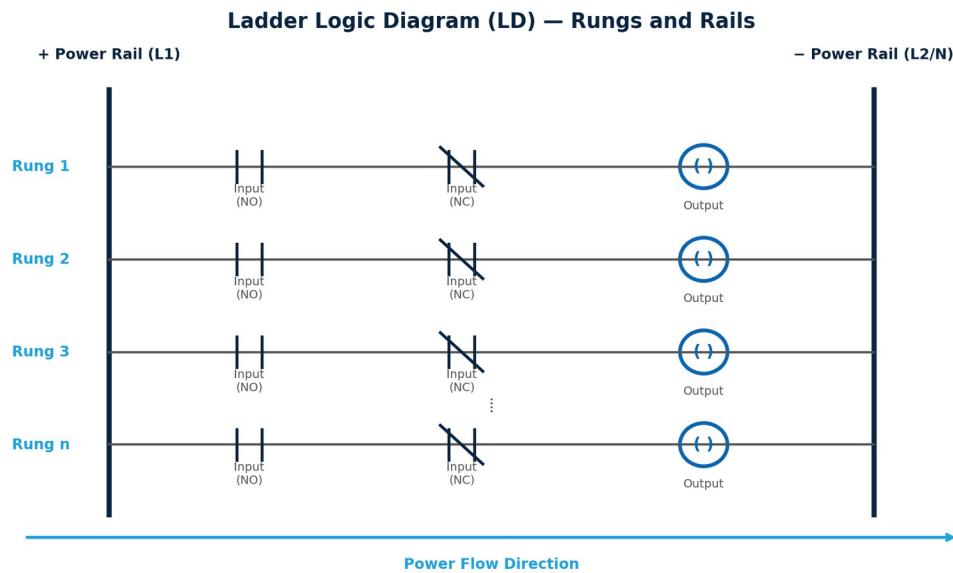


Figure 8.1 — Basic structure of a ladder logic diagram showing rails, rungs, contacts, and output coils.

8.1 Key Symbols

PLC Switching Contact Symbols

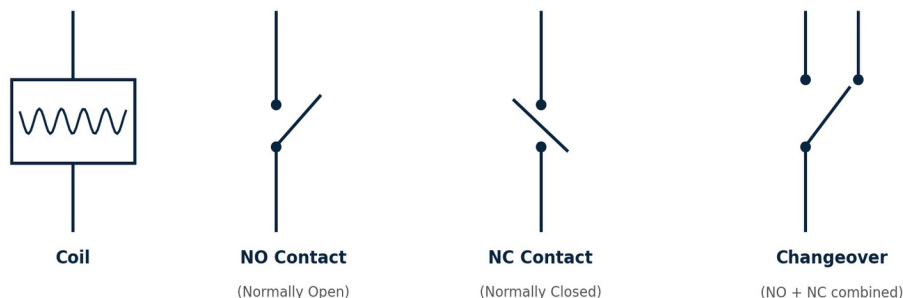


Figure 8.2 — Standard symbols used in ladder logic: coil, NO contact, NC contact, and changeover.

- **Normally Open (NO) contact** — Reads the state of an input. Passes power when the referenced bit is TRUE (1).



- **Normally Closed (NC) contact** — Reverse of NO. Passes power when the referenced bit is FALSE (0).
- **Output Coil** — Writes a value to a bit. Energized (1) when the rung has continuous power flow from left to right.
- **Latching Coil** — Output stays ON even after the rung loses power, until specifically reset.

8.2 A simple worked example — Motor Start/Stop

Imagine a motor that should start when a Start pushbutton is pressed and stop when a Stop pushbutton is pressed. The motor should keep running after the Start button is released, until Stop is pressed. This is the classic seal-in (or latching) circuit.

In ladder logic, the rung looks like this: in series, an NO contact from the Start button is in parallel with an NO contact from the motor's own output bit (this is the seal-in branch). Then in series with an NC contact from the Stop button. The output is the motor coil.

When Start is pressed, power flows to the motor coil. The motor energizes, and the motor's own NO contact closes — sealing the circuit. Now even if Start is released, the seal-in keeps power flowing. When Stop is pressed, its NC contact opens, breaking the circuit. The motor stops, the seal-in opens, and the rung is ready for the next start.

The most important ladder logic habit

Always include a wired Emergency Stop and a fail-safe Stop button using normally-closed contacts in the physical wiring — never rely on software alone for safety. The PLC should see the Stop signal regardless of whether the program is running correctly.

9. Timers, Counters & Math Instructions

Beyond contacts and coils, every PLC supports a rich library of instructions for timing, counting, and arithmetic. These are the workhorses of real machine logic.

9.1 Timers

A timer generates a delay between an input condition becoming true and an output being energized (or de-energized). Three types are standard:

- **TON — On-Delay Timer.** Output turns ON after the input has been ON for the preset time.
- **TOF — Off-Delay Timer.** Output turns OFF after the input has been OFF for the preset time.
- **RTO — Retentive On-Delay Timer.** Like TON, but it remembers accumulated time even if the input briefly turns off. Must be explicitly reset.

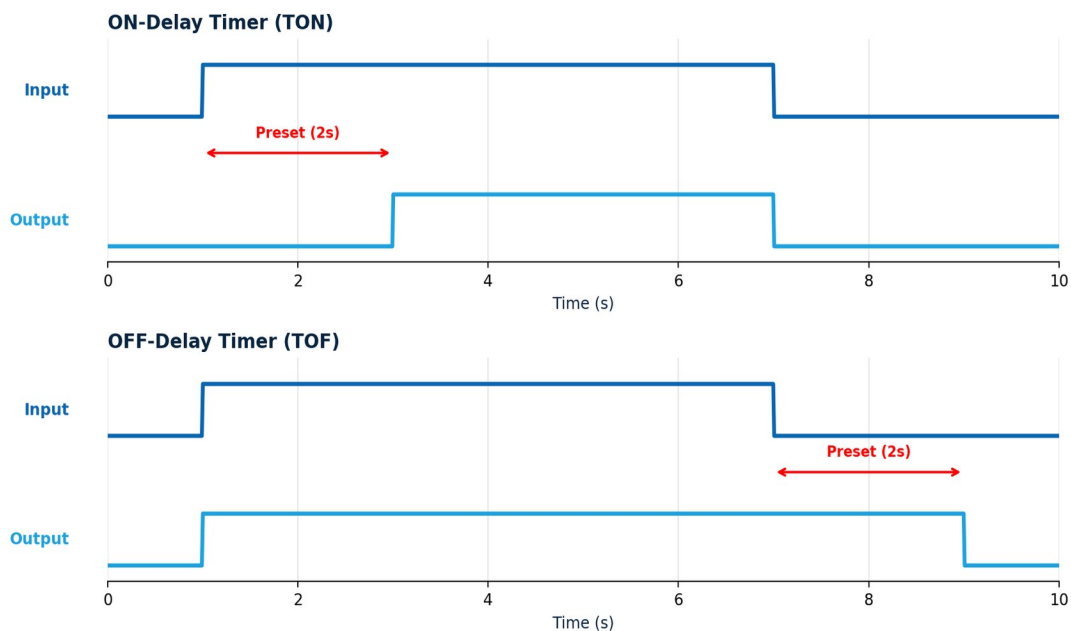


Figure 9.1 — Timing diagrams comparing ON-delay and OFF-delay behavior.

Every timer has three key values: Preset (the target time), Accumulated (the elapsed time), and Status bits (DN = done, EN = enabled, TT = timer timing).

9.2 Counters

Counters track how many times an event has occurred. Three types are standard:

- **CTU — Count Up.** Increments accumulator on each rising edge of the input.
- **CTD — Count Down.** Decrements accumulator on each rising edge of the input.
- **CTUD — Count Up/Down.** Combines both, with separate inputs for up and down counts.



Typical applications include counting products on a conveyor, tracking machine cycles for maintenance scheduling, or counting parts in a batch.

9.3 Math and Comparison Instructions

PLCs handle arithmetic on integer and floating-point variables. Common instructions include ADD, SUB, MUL, DIV (basic arithmetic), MOV (move a value from one location to another), and the comparison family EQU, NEQ, GRT, LES, GEQ, LEQ (equal, not equal, greater than, less than, greater or equal, less or equal). These instructions are essential for analog control, scaling sensor values, and decision making.

9.4 Special instructions

- **MCR — Master Control Relay.** Disables a section of the program. Used for emergency conditions or mode changes.
- **JMP / LBL — Jump and Label.** Skip portions of the program based on a condition.
- **SCP / SCL — Scale.** Convert raw analog values (e.g. 0-32767) into engineering units (e.g. 0-100 °C).
- **PID — Proportional-Integral-Derivative.** Closed-loop control for analog processes like temperature, pressure, or flow.

10. PLC I/O — Sinking and Sourcing

When wiring DC inputs and outputs on a PLC, you must understand the concepts of sinking and sourcing. Getting this wrong is one of the most common wiring mistakes in the field — and it can prevent the PLC from reading sensors correctly.

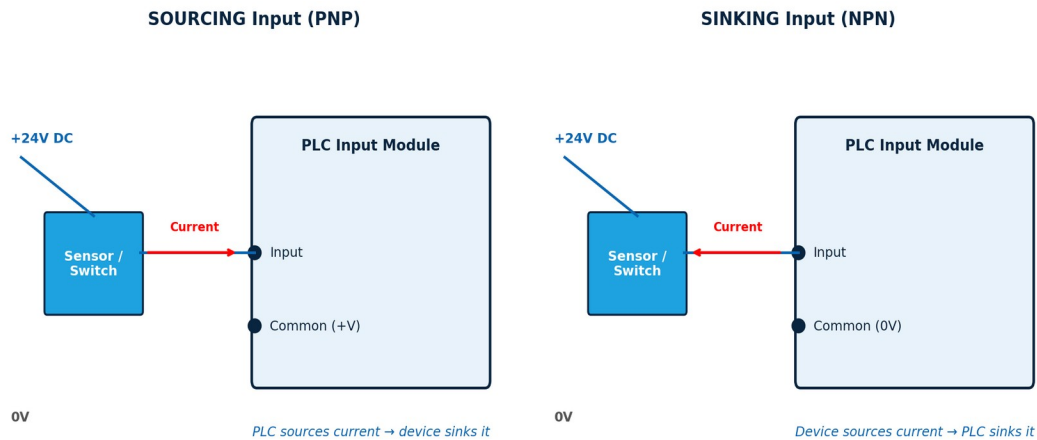


Figure 10.1 — Sourcing input (PNP) and sinking input (NPN) wiring configurations.

10.1 Sourcing (PNP)

In a sourcing arrangement, the device — for example a PNP-type proximity sensor — supplies (sources) positive current into the PLC input terminal when activated. The PLC's input common is connected to 0 V. PNP sensors are the standard in most of Europe and increasingly worldwide.

10.2 Sinking (NPN)

In a sinking arrangement, the device — an NPN-type sensor — pulls current down to 0 V through the PLC input terminal. The PLC's input common is connected to +24 V. NPN sensors are still common in Asian-built equipment.

Practical rule

Always check the PLC input module's polarity (PNP or NPN) before wiring sensors. If you connect a PNP sensor to an NPN-only input — or the reverse — the input will simply never turn on, and you will spend hours wondering why. Mixed-polarity modules exist but must be configured correctly.

11. Sensors Used with PLCs



A PLC is only as good as the sensors it reads from. Different applications demand different sensing technologies, and a competent PLC engineer should know which to specify.

Sensor Type	What it Detects	Typical Application
Inductive Proximity	Metallic objects without contact	Position of metal parts, robot arm limits
Capacitive Proximity	Metals, plastics, liquids, powders	Tank level, plastic part detection
Photoelectric (Optical)	Any object that breaks a light beam	Conveyor item counting, gate safety
Ultrasonic	Distance via sound waves	Tank level, transparent object detection
Limit Switch	Mechanical contact with object	End-of-travel detection on cylinders
RTD / Thermocouple	Temperature	Furnace, oven, motor winding temperature
Pressure Transmitter	Fluid or gas pressure	Hydraulic systems, HVAC, process control
Flow Meter	Liquid or gas flow rate	Water treatment, dosing, batching
Load Cell	Force or weight	Tank weighing, conveyor batching
Encoder	Position or speed of rotation	Servo motors, conveyor position tracking

Sensors output either a discrete signal (on/off, sent to a DI module) or an analog signal — typically 4-20 mA current loop or 0-10 V — that is read by an AI module and scaled to engineering units in the program.



12. Communication Protocols

Modern industrial automation is networked. A PLC almost never works alone — it talks to HMIs, SCADA servers, variable frequency drives, robots, vision systems, and other PLCs. Each conversation happens over a communication protocol, which defines the rules of data exchange.

12.1 Most common industrial protocols

- **Modbus RTU / Modbus TCP** — Simple, open, ubiquitous. RTU over RS-485 serial, TCP over Ethernet. A great starting protocol.
- **Profibus DP** — Siemens-led fieldbus, very common in process industries. Master-slave architecture.
- **Profinet** — Modern Ethernet-based successor to Profibus. Dominant in new Siemens installations.
- **EtherNet/IP** — Rockwell Automation's Ethernet-based protocol. Dominant in North America and on Allen-Bradley systems.
- **EtherCAT** — High-speed deterministic Ethernet, popular in motion control and Beckhoff systems.
- **DeviceNet / ControlNet** — Legacy Rockwell fieldbuses, still encountered in older plants.
- **CAN open** — Common in machine building, especially in Europe.
- **HART** — Used in process instrumentation; carries digital data on top of a 4-20 mA analog signal.
- **AS-Interface (AS-i)** — Simple two-wire bus for connecting sensors and actuators at field level.
- **OPC UA** — Modern, secure, vendor-neutral protocol for connecting PLCs to MES, ERP, and cloud systems. The standard for Industry 4.0.

12.2 Choosing the right protocol

Choice of protocol depends on speed requirements, the equipment ecosystem you are already in, distance, and noise tolerance. Profinet and EtherNet/IP are the dominant choices in new installations. For connecting to legacy equipment, Modbus is almost always supported. For exchanging data with IT systems and the cloud, OPC UA is the modern answer.

Practical tip for engineers

When buying a new sensor, drive, or HMI, always check which protocols it supports and make sure they match your PLC. A mismatch can mean adding a protocol gateway — a costly extra component that also adds a point of failure.

13. Major PLC Brands & Manufacturers



The industrial automation market is dominated by a handful of large manufacturers, each with their own software, hardware, and ecosystem. Learning one platform deeply is more valuable than learning all of them shallowly — but knowing which brand dominates which region helps with job opportunities.

Brand	Country of Origin	Notable Products / Strengths
Siemens	Germany	SIMATIC S7-1200, S7-1500. Dominant globally and in Europe.
Rockwell (Allen-Bradley)	USA	ControlLogix, CompactLogix. Dominant in North America.
Mitsubishi Electric	Japan	MELSEC FX, Q, iQ-R series. Popular in Asia.
Schneider Electric (Modicon)	France	Modicon M340, M580. Inventor of the original PLC.
ABB	Switzerland/Sweden	AC500 series. Strong in process and power industries.
Omron	Japan	Sysmac, CJ, CP series. Popular in packaging and OEM.
Delta	Taiwan	Affordable DVP series, growing global market share.
Beckhoff	Germany	PC-based control, EtherCAT. Strong in motion and machine building.
B&R Automation	Austria	Now part of ABB. Strong in food, packaging, plastics.
LS Electric	South Korea	XGB, XGI series. Popular in Korean industry.
GE Automation / Emerson	USA	RX3i series. Process and energy industries.
Hitachi, Toshiba, Keyence	Japan	Specialty PLCs and vision/automation integration.



14. Applications of PLCs in Industry

PLCs are everywhere in modern industry. Anywhere you find a machine that must be controlled reliably and reprogrammed easily, you find a PLC. Below is a tour of the major application areas.

Manufacturing

- **Automotive** — Assembly lines, robotic welding, paint shops, body-in-white stations. (This is exactly the environment Awet works in at ASAN.)
- **Packaging** — Filling, sealing, labeling, palletizing.
- **Textile** — Spinning, weaving, dyeing process control.
- **Food & Beverage** — Mixing, pasteurization, bottling, batch control.

Process industries

- **Oil & Gas** — Compressor stations, pipelines, refineries (often combined with DCS).
- **Chemical & Pharmaceutical** — Reactor control, batch sequencing.
- **Water & Wastewater** — Pumping stations, filtration, chlorination, SCADA-monitored networks.
- **Power generation** — Boiler control, turbine auxiliaries, substation automation.

Building and infrastructure

- **HVAC** — Heating, ventilation, and air conditioning control in large buildings.
- **Elevators & escalators** — Motion sequencing and safety interlocks.
- **Traffic systems** — Signal control, tunnel ventilation, smart highway sensors.

Other

- **Mining & metals** — Crushers, conveyors, smelters.
- **Agriculture** — Greenhouse climate, irrigation, grain handling.
- **Renewable energy** — Solar plant tracking, wind turbine pitch and yaw control.

15. Advantages & Disadvantages of PLCs

PLCs have transformed industrial control, but they are not the right choice for every application. Knowing both sides helps you make better design decisions.

15.1 Advantages

- **Flexible** — Logic changes are software changes; no rewiring required.
- **Reliable** — Designed for 24/7 operation in harsh industrial environments.
- **Compact** — A small PLC can replace a large bank of relays.



-
- **Fast** — Scan times in the millisecond range, allowing fast machine response.
 - **Modular** — Add modules as the system grows.
 - **Networkable** — Can communicate with HMIs, SCADA, drives, and other PLCs.
 - **Diagnosable** — Built-in fault detection makes troubleshooting much easier than with relay panels.
 - **Standardized** — IEC 61131-3 provides a common framework across brands.
 - **Cost-effective** — For mid-size and large systems, far cheaper than equivalent relay logic.

15.2 Disadvantages

- **Initial cost** — For very small applications (e.g. one or two relays), a PLC can be overkill.
- **Vendor lock-in** — Each brand has its own software and proprietary features.
- **Environmental limits** — Even rugged PLCs degrade in extreme heat, humidity, or vibration without proper enclosures.
- **Specialist knowledge** — Requires trained personnel to program and maintain.
- **Cyber security** — Networked PLCs are now attack surfaces and require proper IT/OT security.
- **Limited motion control** — For very high-speed motion, dedicated motion controllers may be needed.



16. Safety, Redundancy & Best Practices

Industrial systems control real machines that can hurt people. A professional PLC engineer treats safety as the first priority, not the last.

16.1 Safety principles

- **Hardware Emergency Stops** — Must be hard-wired to remove power directly from actuators, not just signal the PLC. Standards like ISO 13849 and IEC 62061 define the required performance levels.
- **Safety PLCs** — For machines with significant risk, use a dedicated Safety PLC (e.g. Siemens S7-1500F, AB GuardLogix) certified to SIL or PL standards.
- **Light curtains and safety mats** — Hard-wired through safety relays or safety PLCs, never through standard inputs.
- **Lockout / Tagout (LOTO)** — Always isolate electrical and pneumatic energy before maintenance, regardless of PLC state.

16.2 Redundancy

In processes where downtime is unacceptable — refineries, power plants, large data centers — engineers use redundant PLC systems. A backup CPU continuously mirrors the primary CPU's state, and takes over within milliseconds if the primary fails. Common redundancy levels are CPU redundancy, I/O redundancy, power supply redundancy, and network redundancy.

16.3 Programming best practices

- **Comment everything** — Future you (or the next engineer) will be grateful.
- **Use meaningful tag names** — Motor_Start, not B3:0/1.
- **Organize into routines** — One routine per machine area or function.
- **Version your programs** — Save dated backups before every modification.
- **Document I/O wiring** — Maintain accurate wiring diagrams in sync with the PLC program.
- **Test before going live** — Use simulation mode where possible.

17. Troubleshooting & Maintenance

Even the most reliable systems eventually fail. A good PLC technician has a systematic troubleshooting process that finds the root cause quickly without guessing.

17.1 The systematic approach

- **1. Observe** — What exactly is not working? Get the full picture before touching anything.



- **2. Check status LEDs** — CPU run, power, input, output indicators reveal a huge amount.
- **3. Read the diagnostic buffer** — Modern PLCs log faults with timestamps. Read it first.
- **4. Force inputs / monitor logic** — In online mode, watch which rungs are live and which aren't.
- **5. Test physical I/O** — Use a multimeter to verify sensors and actuators are wired and powered.
- **6. Isolate** — Halve the problem: is it the field device, the wiring, the input card, or the logic?
- **7. Fix and document** — When solved, write down what was wrong and how you fixed it.

17.2 Preventive maintenance

- **Check enclosure cooling** — Cabinets overheating shortens PLC life. Clean filters, replace failing fans.
- **Inspect connections** — Vibration loosens terminals over time. Re-torque on schedule.
- **Back up programs regularly** — And store backups offsite or in cloud, not just on the laptop in the workshop.
- **Replace batteries** — On older PLCs, memory backup batteries die after 3–7 years.
- **Update firmware cautiously** — Test on a non-production system first; document the version before and after.

18. Career Pathways in PLC & Automation

PLC skills open doors to a wide range of careers across virtually every industry. Below are the main roles you can target as you build experience.

Role	What They Do	Typical Background
PLC / Automation Engineer	Designs, programs, commissions PLC systems	Electrical, mechatronics, mechanical eng.
Controls Technician	Maintains, troubleshoots PLC equipment	Technical diploma, on-the-job training
SCADA Engineer	Designs supervisory systems above the PLC	PLC + IT background
Robotics Engineer	Integrates PLCs with industrial robots	Mechatronics, robotics
MES / IIoT Engineer	Connects PLCs to enterprise & cloud systems	PLC + databases, networking
Systems Integrator	Combines PLCs, drives, sensors, software into solutions for clients	Broad automation experience



Role	What They Do	Typical Background
Field Service Engineer	Travels to plants for commissioning and repair	PLC + mechanical + diagnostic skills
Functional Safety Engineer	Designs and certifies safety systems	PLC + ISO 13849 / IEC 61508 expertise

A note for engineers in Africa and beyond

PLC and automation skills travel extremely well. Engineers from Ethiopia, Korea, Vietnam, Mexico, and the Philippines work in plants all over the world precisely because the standards (Siemens, Rockwell, IEC 61131) are global. Build a portfolio of completed projects and you become hireable everywhere.

19. Learning Roadmap & Resources

Becoming proficient in PLCs is a journey of roughly one to two years of focused part-time study and practice. Below is a recommended path.

Stage 1 — Foundations (4–8 weeks)

Basic electrical theory: voltage, current, resistance, AC vs DC, relays, contactors. Read electrical schematics. Understand sensors and basic instrumentation.

Stage 2 — PLC Fundamentals (6–10 weeks)

Learn ladder logic on a single platform — Siemens TIA Portal (free trial) or the free Logix Designer Lite. Master contacts, coils, timers, counters, and the basic math instructions.

Stage 3 — Practical Projects (8–12 weeks)

Build simulated projects: traffic light, motor start-stop with seal-in, conveyor sorter, batch process. Use free simulators like Factory I/O, Siemens PLCSIM, or open-source OpenPLC.

Stage 4 — HMI & Communications (6–8 weeks)

Add an HMI to your projects (WinCC, FactoryTalk View ME, or open alternatives like AdvancedHMI). Learn Modbus, then Profinet or EtherNet/IP.

Stage 5 — Advanced Topics (ongoing)

PID control loops, motion control, SCADA, safety PLCs, OPC UA, and the basics of Industry 4.0. Build a public portfolio of projects on GitHub or YouTube.



Recommended resources

- **Books** — Petruzella's Programmable Logic Controllers, Hugh Jack's Automating Manufacturing Systems.
- **YouTube** — RealPars, PLC Professor, PLC Academy, GeeksTechCorner.
- **Courses** — Udemy (Paul Lynn), Coursera, Inductive University (free, Ignition SCADA), AYE TechEngine.
- **Software (free or trial)** — Siemens TIA Portal V18 Trial, Logix Designer Lite, OpenPLC.
- **Simulators** — Factory I/O, Siemens PLCSIM, Allen-Bradley Emulate.



20. Final Words from AYE Tech Hub

Programmable Logic Controllers are at the heart of every modern factory, plant, and automated building. Mastering them is one of the most practical investments an engineer can make today — the skills transfer across industries, across countries, and across decades. A PLC engineer who started learning in 1985 still uses many of the same fundamentals today, only with faster hardware and better software.

This course gives you the structured foundation. The next step is yours: install a free simulator this week, build your first ladder logic program, and start a habit of one small project per month. Document each one publicly — on LinkedIn, on YouTube, on a personal blog. Within a year, your portfolio will speak louder than any certificate.

AYE Tech Hub's mission is to make modern engineering education accessible in Tigrinya, Amharic, and English. We will continue to publish course guides, video tutorials, and practical projects that engineers anywhere in the world can use to grow their careers. We are with you on the journey.

Stay connected

Visit ayetechub.com for more courses, tools, and engineering resources. Join us on YouTube, Telegram, LinkedIn, Instagram, TikTok, Threads, and Facebook (@ayetechub / @ayetechpro / ayetecheengine). For services and direct contact, email awetgknway@gmail.com.



Engineering the Future through Technology & Knowledge

— End of Course Guide —